

LightTracking Setup Guide

Best-practice setup for a 10 m stage and fast-moving performers

Target outcome

- Smooth follow on a running performer across 10 m
- Low tracking latency (aim: < 150 ms end-to-end)
- Stable tracking in the full performance area

System summary

Tags send UWB ranges to anchors
Anchors publish batches to MQTT
Tracking server solves position (25 Hz target)
DMX engine updates fixtures (30 Hz)

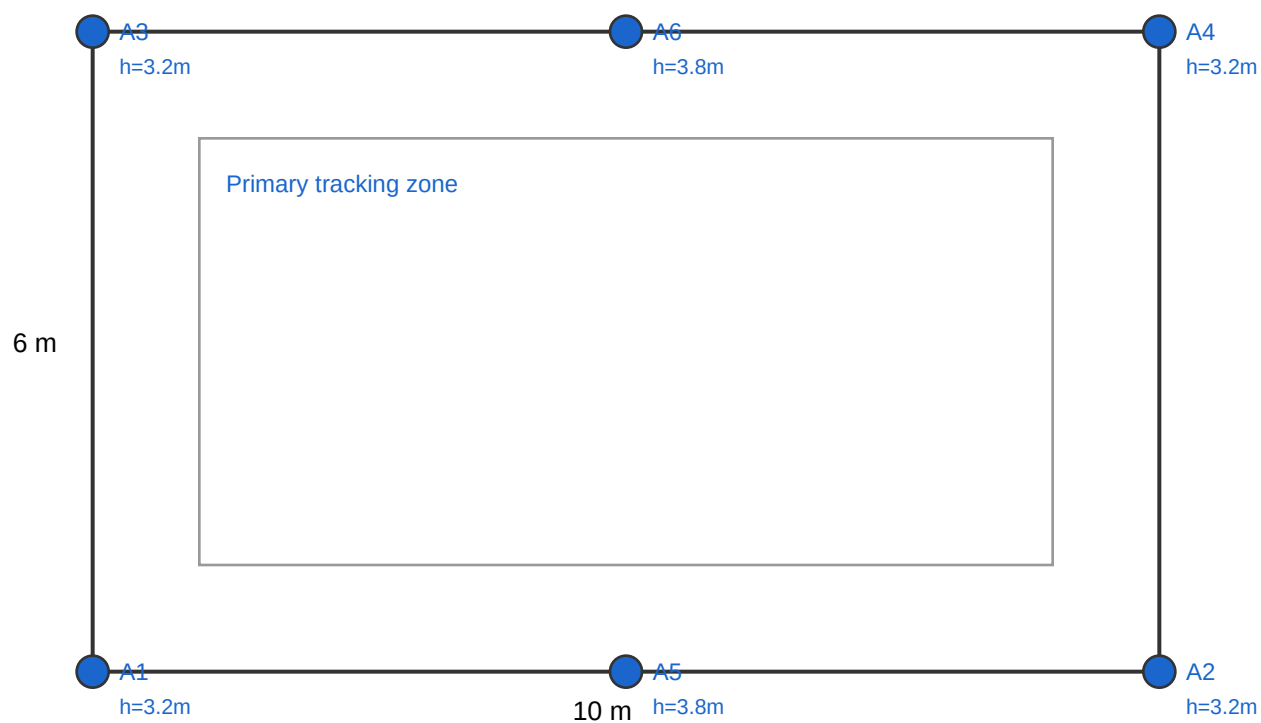
Quick recommended settings

Anchor batch_period_ms: 50 (start here)
tracking_hz: 25 (20-30 range)
stale_timeout_ms: 500
lost_timeout_ms: 1500

Note: If the stage is larger or RF is noisy, increase anchor count and keep line-of-sight.

Physical setup: anchor placement

Example for 10 m wide x 6 m deep stage (top-down view)



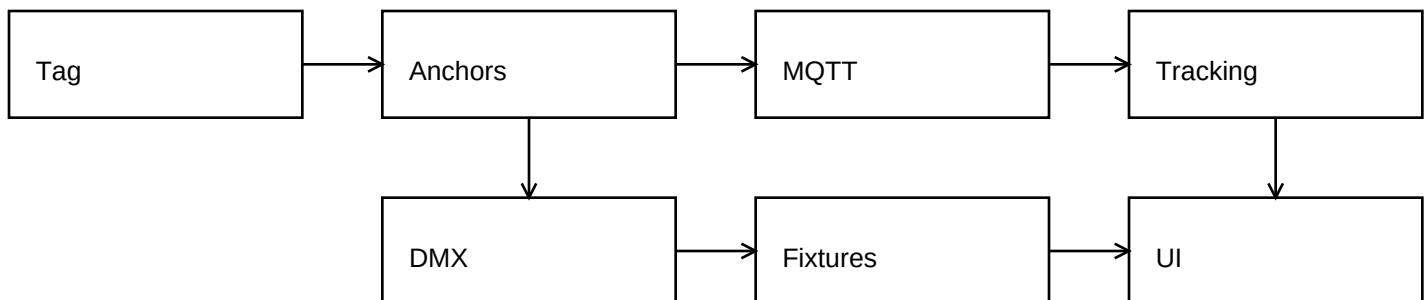
Placement tips:

- Keep anchors high (3-4 m) with clear line-of-sight
- Surround the stage perimeter; avoid all anchors on one line
- Use at least 6 anchors for fast motion

Backend settings & signal flow

Setting	Recommended	Notes
batch_period_ms	50	Anchor setting; 80-100 if noisy
tracking_hz	25	Tracking tick rate
stale_timeout_ms	500	STALE threshold
lost_timeout_ms	1500	LOST threshold

Signal flow (simplified)



Apply settings via backend API:

- `/api/v1/devices/{mac}/apply-settings` for `batch_period_ms`
- `/api/v1/settings` key = `rates.global` for `tracking_hz`